ModelIt!

Teacher's Guide

Ecosystem Engineers

How One Species Creates a World

Quick Reference

Grade Level	Grade NE
NGSS Standard	5-LS2-1, 5-ESS2-1, 3-5-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-LS2-1, 5-ESS2-1, 3-5-ETS1-3

Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment. Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and atmosphere interact. Plan and carry out fair tests in which variables are controlled and failure points are considered.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a 7-component computational model showing cascading relationships from beaver activity through dam construction, water management, wetland creation, to biodiversity support.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems** — Multiple species depend on the habitat created by beavers — demonstrating how organisms in an ecosystem are interconnected through their physical environment.
Crosscutting Concept	**Systems and System Models** — A system can be described in terms of its components and their interactions — the 7-component model reveals how changing one input cascades through the entire system.

CAST Testing Alignment

MULTIPLE CHOICE:

In a watershed with beavers, a logging company removes most of the nearby trees. According to the cascade model, which component will be affected FIRST? A) Biodiversity B) Dam Size C) Wetland Area D) Water Level

CONSTRUCTED RESPONSE:

Using your 7-component model, explain what happens to the entire ecosystem over time when beavers are removed from a watershed. Describe each step in the cascade from dam decay to biodiversity loss.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson builds on Level 1 by expanding the simple beaver-dam-pond model into a full ecosystem cascade. Students now see how the basic building process creates a chain of effects that extends far beyond the dam itself.

From Dam to Ecosystem

A beaver dam does not just create a pond — it creates an entirely new type of habitat. The dam slows water flow, raises the water table, floods adjacent land, creates deep-water zones, generates edge habitat where water meets land, and increases the overall complexity of the landscape. Each of these changes supports different species.

The Wetland Effect

Beaver-created wetlands are among the most biologically productive ecosystems on Earth. They support amphibians that need still water for breeding, fish that need deep pools for refuge, birds that need shoreline for nesting, insects that need standing water for larvae, and plants that need waterlogged soil. A single beaver pond can increase local bird species by 75%.

Water Management

Beavers are remarkably effective water managers. Their dams slow flood waters, reducing downstream flooding by 20-60%. They recharge groundwater aquifers by spreading water across floodplains. They filter sediment and pollutants from water as it passes through their ponds. And they store water during wet periods and release it during dry periods, buffering against both floods and droughts.

The Removal Cascade

When beavers are removed (by trapping, habitat destruction, or predation), the cascade reverses. Dams decay within 1-3 years without maintenance. Ponds drain, lowering the water table. Wetlands dry out, killing aquatic vegetation. Fish, amphibians, and invertebrates lose habitat. Birds and mammals that depended on the wetland must relocate or die. The rich ecosystem reverts to a simple stream channel.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Beaver Population, Tree Availability, and Rainfall as external components. Dam Size, Water Level, Wetland Area, and Biodiversity are internal components forming a cascade chain.

E **Establish Relationships**

Students trace the cascade: Beaver Population + Tree Availability → Dam Size → Water Level (also boosted by Rainfall) → Wetland Area → Biodiversity. Each link is positive.

V **Visualize & Model**

Students build the full 7-component model with three external inputs feeding a four-step cascade of internal components.

E **Evaluate Outcomes**

Students run thriving, drought, removal, and deforestation scenarios to test different parts of the cascade.

R **Revise & Extend**

Students add Water Temperature, Sediment Filtering, or Predator Population to further explore ecosystem complexity.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Remember our Level 1 model? Beavers, trees, dam, pond. Today we go FURTHER — what happens AFTER the pond forms?"*

Watch For

Show aerial drone footage of beaver wetlands. Ask: How many different species can you spot?

Slide 2: Learning Objectives

3 min

Say: *"We are expanding from 4 components to 7. This is a BIG model — but you are ready."*

Watch For

Preview the new vocabulary: cascade, wetland, hydrosphere. Connect to Level 1 knowledge.

Slide 3: Big Question

3 min

Say: *"A beaver family is usually 4-8 animals. How can 8 beavers support HUNDREDS of other species?"*

Watch For

Quick brainstorm: students list species that might live in a beaver wetland. Aim for 20+.

Slide 4: LEVER Framework

2 min

Say: *"Our model today has a CHAIN — each component feeds the next. This is called a cascade."*

Watch For

Draw the cascade as a staircase: beaver → dam → water → wetland → biodiversity. Each step higher.

Slide 5: Activity 1: Components

8 min

Say: *"Three things WE can control, four things the SYSTEM produces. Let us sort them."*

Watch For

Guide sorting of all 7 components. Emphasize: Level 1 had 4. Level 2 has 7. More components = more realistic.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the chain. Beavers + Trees feed the Dam. The Dam feeds Water Level. Rainfall ALSO feeds Water Level. Water feeds Wetland. Wetland feeds Biodiversity."*

Watch For

Build the cascade arrow by arrow. Key moment: Rainfall feeds Water Level directly, not through the dam.

Slide 7: Activity 3: Simulation

12 min

Say: *"Four tests today. Thriving colony, drought, beavers removed, and deforestation. Predict each one!"*

Watch For

Run scenarios in order. After each, pause to compare predictions to results. Drought is the most surprising.

Slide 8: Discoveries

4 min

Say: *"One family of beavers, hundreds of species supported. That is the power of ecosystem engineering."*

Watch For

Summarize cascade findings. Compare: What did Level 1 teach us? What did Level 2 ADD?

Slide 9: STEM Challenge

12 min

Say: *"Now paint what you modeled — three stages, scientifically accurate, and beautiful!"*

Watch For

Distribute art supplies. Groups plan panels, assign species, and begin painting.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Beaver Population, Tree Availability, Rainfall
- **INTERNAL:** Dam Size, Water Level, Wetland Area, Biodiversity

Relationships

- **Beaver Population to Dam Size:** POSITIVE (+) — More beavers means more workers to cut trees, haul branches, and pack mud. Larger families build bigger, more complex dam systems.
- **Tree Availability to Dam Size:** POSITIVE (+) — More trees means more building material. Even large beaver families are limited by material availability.
- **Dam Size to Water Level:** POSITIVE (+) — Bigger dams hold back more water, raising the water level higher behind the dam and flooding a larger area.
- **Rainfall to Water Level:** POSITIVE (+) — More rainfall fills the pond faster and maintains higher water levels. This input feeds Water Level directly, bypassing the dam.
- **Water Level to Wetland Area:** POSITIVE (+) — Higher water levels spread across more land, creating more wetland habitat. Water seeps into surrounding soil, raising the water table.
- **Wetland Area to Biodiversity:** POSITIVE (+) — Larger wetlands support more diverse habitat types (deep water, shallow water, mudflat, shoreline), each attracting different species.

Key Discoveries

- Beaver activity creates a CASCADE: dam → water → wetland → biodiversity — each step depends on the one before it
- Three external inputs ALL matter — losing any one significantly weakens the system
- Beavers can buffer against drought because their dams store water — but they cannot create water from nothing
- Removing beavers doesn't just affect the pond — it collapses the ENTIRE chain of dependent species
- The beaver-created wetland supports more species than the original stream ever could

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Beaver dams are permanent structures"	Beaver dams require constant maintenance. Without beavers actively repairing them, dams begin decaying within months. Water pressure, storms, and ice damage all wear on the dam. This is why beaver REMOVAL leads to dam COLLAPSE — no maintenance means no dam.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Any water creates a wetland"	A puddle is not a wetland. Wetlands require sustained water saturation that changes soil chemistry, supports specific plant communities, and provides habitat structure. Beaver dams create the sustained, managed water levels that true wetlands need — a one-time flood does not.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"More rainfall is always better for wetlands"	While wetlands need water, TOO much rainfall can actually cause problems: flooding the dam, washing away dam material, and creating fast-flowing water that destroys habitat structure. Beaver dams work best with moderate, consistent rainfall. Extreme events can overwhelm the system.